Journal of Energy, Heat and Mass Transfer, vol. 32, pp. 243-263, 2010

Radiation Effects on MHD Flow of a Chemically Reacting Fluid Past a Vertical Plate with Viscous Dissipation

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Abstract

This paper is focused on the effect of thermal radiation on the natural convective heat and mass transfer flow of a viscous, incompressible, electrically conducting and dissipating fluid past an impulsively started moving vertical plate adjacent to porous regime, taking into account of the homogeneous chemical reaction of first order. The dimensionless governing equations are solved using a Regular Perturbation method. A parametric study is performed to illustrate the influence of thermo physical parameters on the velocity, temperature and concentration profiles. Also, the skin-friction, Nusselt number and Sherwood numbers are computed. It is found that thermal radiation reduces both the velocity and temperature in the boundary layer. Increasing permeability enhances the flow velocity. An increase in the viscous dissipation causes a rise in both the velocity and temperature. This model finds applications in solar energy collection systems, porous combustors and transport in fires in porous media (forest fires).

INTRODUCTION

The interaction of heat and mass transfer with magnetic field has attracted the interest of many researches in view of its applications in MHD generators, plasma studies, nuclear reactors, geophysics and astrophysics. Elbashbeshy [5] studied heat and mass transfer along a vertical plate with variable surface temperature and concentration in the presence of magnetic field. Shankar and Kishan [13] discussed the effect of mass transfer on the MHD flow past an impulsively started vertical plate with variable temperature or constant heat flux.

For some industrial applications such as glass production and furnace design in space technology applications, cosmical flight aerodynamics rocket, propulsion systems, plasma physics which operate at higher temperatures, radiation effects can be significant. Chaudhary *et al.* [4] studied the radiation effects with simultaneous thermal and mass diffusion in MHD mixed convection flow from a vertical surface. Ramachandra Prasad *et al.* [9] studied the transient

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